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Inventor(s)	Kawasaki; Koichiro

Acoustic wave device

Abstract

A multiplexer includes first and second filters, a resin layer, and an electrically conductive layer. A pass band of the first filter is different from a pass band of the second filter. The first filter is in contact with an electrically conductive layer, and the second filter is not in contact with the electrically conductive layer. The second filter has a greater thermal resistance value than the first filter, and lowers the pass band in response to a temperature rise. A region of the piezoelectric substrate closer to the first filter is a first region, and a region of the piezoelectric substrate farther from the first filter is a second region. Areas in the first and second regions where an acoustic wave resonator included in a parallel-arm circuit is provided are first and second areas. The second area is greater than the first area.

Inventors:	Kawasaki; Koichiro (Nagaokakyo, JP)
Applicant:	Murata Manufacturing Co., Ltd. (Nagaokakyo, JP)
Family ID:	82654481
Assignee:	MURATA MANUFACTURING CO., LTD. (Kyoto, JP)
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Primary Examiner: Outten; Samuel S

Attorney, Agent or Firm: Keating & Bennett, LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority to Japanese Patent Application No. 2021-014347 filed on Feb. 1, 2021 and is a Continuation Application of PCT Application No. PCT/JP2021/048243 filed on Dec. 24, 2021. The entire contents of each application are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to an acoustic wave device including an acoustic wave element, and more specifically, to a technology to reduce degradation of characteristics of a multiplexer including multiple acoustic wave elements due to a temperature rise.

2. Description of the Related Art

(2) International Publication No. 2013/146374 discloses a structure including multiple acoustic wave elements mounted on a mount board by flip chip bonding. In the acoustic wave device disclosed in International Publication No. 2013/146374, the acoustic wave elements on the mount board are sealed with resin, and a support board forming the acoustic wave elements is disposed in direct contact with a highly thermally conductive member. This structure can immediately dissipate heat generated by the acoustic wave elements to the outside via the highly thermally conductive member. Thus, the acoustic wave device can improve the heat dissipation performance, and reduce degradation of the characteristics of the acoustic wave device.

(3) In such a structure as that in International Publication No. 2013/146374, heat generated by one acoustic wave element may be transferred to another acoustic wave element via the sealing resin and the highly thermally conductive member.

(4) The material of the piezoelectric substrate forming the acoustic wave elements may vary between the acoustic wave elements. In this case, the acoustic wave elements may have different temperature characteristics. When heat generated by one acoustic wave element is transferred to a different acoustic wave element via the sealing resin or the highly thermally conductive member for heat dissipation, the characteristics of the different acoustic wave element may be degraded.

SUMMARY OF THE INVENTION

(5) Preferred embodiments of the present invention provide acoustic wave devices each including multiple acoustic wave elements with reduced degradation of characteristics due to heat.

(6) An acoustic wave device according to a preferred embodiment of the present invention includes a mount board, a first filter and a second filter on the mount board, a resin layer sealing the first filter and the second filter, and an electrically conductive layer covering the resin layer. The second filter is adjacent to the first filter in a first direction. The second filter has a pass band higher than a pass band of the first filter. Each of the first filter and the second filter includes a piezoelectric substrate, and an acoustic wave resonator on the piezoelectric substrate. The first filter is in contact with the electrically conductive layer, and the second filter is not in contact with the electrically conductive layer. The second filter is a ladder acoustic wave filter including at least one parallel-arm resonator and at least one series-arm resonator, or a longitudinally coupled acoustic wave filter including at least one longitudinally coupled resonator. A thermal resistance value of the piezoelectric substrate in the second filter in the first direction is greater than a thermal resistance value of the piezoelectric substrate in the first filter in the first direction. The second filter has characteristics lowering the pass band in response to a temperature rise of the piezoelectric substrate of the second filter. With respect to a virtual line perpendicular or substantially perpendicular to the first direction and passing through a center of the piezoelectric substrate of the second filter, a region of the piezoelectric substrate closer to the first filter is defined as a first region, and a region of the piezoelectric substrate farther from the first filter is defined as a second region. All of the at least one parallel-arm resonator included in the ladder acoustic wave filter is defined as a parallel-arm circuit, all of the at least one series-arm resonator is defined as a series-arm circuit, and all of the at least one longitudinally coupled resonator included in the

longitudinally coupled acoustic wave filter is defined as a longitudinally coupled circuit. Areas in the first region and the second region where the acoustic wave resonator included in the parallel-arm circuit or the longitudinally coupled circuit is provided are defined as a first area and a second area. The second area is greater than the first area.

(7) An acoustic wave device according to a preferred embodiment of the present invention includes a mount board, a first filter and a second filter on the mount board, a resin layer sealing the first filter and the second filter, and an electrically conductive layer covering the resin layer. The second filter is adjacent to the first filter in a first direction. The second filter has a pass band higher than a pass band of the first filter. Each of the first filter and the second filter includes a piezoelectric substrate, and an acoustic wave resonator on the piezoelectric substrate. The first filter is in contact with the electrically conductive layer, and the second filter is not in contact with the electrically conductive layer. The second filter is a ladder acoustic wave filter including at least one parallel-arm resonator and at least one series-arm resonator, or a longitudinally coupled acoustic wave filter including at least one longitudinally coupled resonator. A thermal resistance value of the piezoelectric substrate in the second filter in the first direction is greater than a thermal resistance value of the piezoelectric substrate in the first filter in the first direction. The second filter has characteristics to raise the pass band in response to a temperature rise of the piezoelectric substrate of the second filter. With respect to a virtual line perpendicular or substantially perpendicular to the first direction and passing through a center of the piezoelectric substrate of the second filter, a region of the piezoelectric substrate closer to the first filter is defined as a first region, and a region of the piezoelectric substrate farther from the first filter is defined as a second region. All of the at least one parallel-arm resonator included in the ladder acoustic wave filter is defined as a parallel-arm circuit, all of the at least one series-arm resonator is defined as a series-arm circuit, and all of the at least one longitudinally coupled resonator included in the longitudinally coupled acoustic wave filter is defined as a longitudinally coupled circuit. Areas in the first region and the second region where the acoustic wave resonator included in the parallel-arm circuit or the longitudinally coupled circuit is provided are defined as a first area and a second area. The first area is greater than the second area.

(8) An acoustic wave device according to a preferred embodiment of the present invention includes a mount board, a first filter and a second filter on the mount board, a resin layer sealing the first filter and the second filter, and an electrically conductive layer covering the resin layer. The second filter is adjacent to the first filter in a first direction. The second filter has a pass band lower than a pass band of the first filter. Each of the first filter and the second filter includes a piezoelectric substrate, and an acoustic wave resonator on the piezoelectric substrate. The first filter is in contact with the electrically conductive layer, and the second filter is not in contact with the electrically conductive layer. The second filter is a ladder acoustic wave filter including at least one parallel-arm resonator and at least one series-arm resonator, or a longitudinally coupled acoustic wave filter including at least one longitudinally coupled resonator. A thermal resistance value of the piezoelectric substrate in the second filter in the first direction is greater than a thermal resistance value of the piezoelectric substrate in the first filter in the first direction. The second filter has characteristics to raise the pass band in response to a temperature rise of the piezoelectric substrate of the second filter. With respect to a virtual line perpendicular or substantially perpendicular to the first direction and passing through a center of the piezoelectric substrate of the second filter, a region of the piezoelectric substrate closer to the first filter is defined as a first region, and a region of the piezoelectric substrate farther from the first filter is defined as a second region. All of the at least one parallel-arm resonator included in the ladder acoustic wave filter is defined as a parallel-arm circuit, all of the at least one series-arm resonator is defined as a series-arm circuit, and all of the at least one longitudinally coupled resonator included in the longitudinally coupled acoustic wave filter is defined as a longitudinally coupled circuit. Areas in the first region and the second region where the acoustic wave resonator included in the serial-arm circuit or the longitudinally

coupled circuit is provided are defined as a third area and a fourth area. The fourth area is greater than the third area.

(9) An acoustic wave device according to a preferred embodiment of the present invention includes a mount board, a first filter and a second filter on the mount board, a resin layer sealing the first filter and the second filter, and an electrically conductive layer covering the resin layer. The second filter is adjacent to the first filter in a first direction. The second filter has a pass band lower than a pass band of the first filter. Each of the first filter and the second filter includes a piezoelectric substrate, and an acoustic wave resonator on the piezoelectric substrate. The first filter is in contact with the electrically conductive layer, and the second filter is not in contact with the electrically conductive layer. The second filter is a ladder acoustic wave filter including at least one parallel-arm resonator and at least one series-arm resonator, or a longitudinally coupled acoustic wave filter including at least one longitudinally coupled resonator. A thermal resistance value of the piezoelectric substrate in the second filter in the first direction is greater than a thermal resistance value of the piezoelectric substrate in the first filter in the first direction. The second filter has characteristics to reduce the pass band in response to a temperature rise of the piezoelectric substrate of the second filter. With respect to a virtual line perpendicular or substantially perpendicular to the first direction and passing through a center of the piezoelectric substrate of the second filter, a region of the piezoelectric substrate closer to the first filter is defined as a first region, and a region of the piezoelectric substrate farther from the first filter is defined as a second region. All of the at least one parallel-arm resonator included in the ladder acoustic wave filter is defined as a parallel-arm circuit, all of the at least one series-arm resonator is defined as a series-arm circuit, and all of the at least one longitudinally coupled resonator included in the longitudinally coupled acoustic wave filter is defined as a longitudinally coupled circuit. Areas in the first region and the second region where the acoustic wave resonator included in the serial-arm circuit or the longitudinally coupled circuit is provided are defined as a third area and a fourth area. The third area is greater than the fourth area.

(10) In each of the acoustic wave devices according to preferred embodiments of the present invention, an acoustic wave element (first filter) having a relatively small thermal resistance value is in contact with an electrically conductive layer covering the outer periphery of the acoustic wave element, and an acoustic wave element (second filter) having a relatively large thermal resistance value is not in contact with the electrically conductive layer. In accordance with the temperature characteristics of frequency of the second filter having a relatively large thermal resistance value, the position of the acoustic wave resonator included in the second filter is determined. This structure can reduce or prevent degradation of the characteristics of the acoustic wave devices due to heat.

(11) The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram of an example of a circuit configuration of an acoustic wave device according to a first preferred embodiment of the present invention.

(2) FIG. 2 is cross-sectional view of a multiplexer in FIG. 1.

(3) FIG. 3 is a diagram describing a piezoelectric substrate in detail.

(4) FIG. 4 is a cross-sectional view of a multiplexer according to a comparative example.

(5) FIG. 5 is a diagram of an example of a temperature distribution map in the multiplexer illustrated in FIG. 1.

- (6) FIG. 6 is a graph illustrating effects of the materials of piezoelectric substrates and the contact state between the piezoelectric substrates and an electrically conductive layer on the filter temperature.
- (7) FIG. 7 is a graph describing a loss (parallel-arrangement loss) caused in filters adjacent to each other, changing in response to a temperature rise.
- (8) FIG. 8 is a graph describing a parallel-arrangement loss in case 1.
- (9) FIG. 9 is a diagram describing an arrangement of parallel-arm resonator units in case 1.
- (10) FIGS. 10A and 10B are diagrams describing areas of regions in which acoustic wave resonators are provided.
- (11) FIG. 11 is a graph describing a parallel-arrangement loss in case 2.
- (12) FIG. 12 is a diagram describing an arrangement of parallel-arm resonator units in case 2.
- (13) FIG. 13 is a diagram describing a parallel-arrangement loss in case 3.
- (14) FIG. 14 is a diagram describing an arrangement of series-arm resonator units in case 3.
- (15) FIG. 15 is a diagram describing a parallel-arrangement loss in case 4.
- (16) FIG. 16 is a diagram describing an arrangement of series-arm resonator units in case 4.
- (17) FIG. 17 is an example of a circuit configuration of a multiplexer according to a second preferred embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(18) Preferred embodiments of the present invention are described below in detail with reference to the drawings. Throughout the drawings, the same or equivalent components are denoted with the same reference signs without being redundantly described.

First Preferred Embodiment

(19) Structure of Multiplexer

(20) FIG. 1 is a diagram of a circuit configuration of a multiplexer 10 as an example of an acoustic wave device according to a first preferred embodiment of the present invention. The multiplexer 10 is a filter used for, for example, a transmission/reception circuit in a communication device.

(21) With reference to FIG. 1, the multiplexer 10 includes an antenna terminal T1, and a transmission (TX) filter 100 and a reception (RX) filter 200 electrically connected to an antenna ANT at the antenna terminal T1. An example of the multiplexer 10 in FIG. 1 is a duplexer including two filters.

(22) The filter 100 is a ladder filter connected between the antenna terminal T1 and a transmission terminal T2. The filter 100 filters a signal received at the transmission terminal T2 and outputs the signal from the antenna ANT. The filter 100 allows signals in a pass band BW1 (first pass band) to pass therethrough.

(23) The transmission filter 100 includes a series-arm circuit including series-arm resonator units S1 to S5 serially connected between the antenna terminal T1 and the transmission terminal T2, and a parallel-arm circuit including parallel-arm resonator units P1 to P4 connected between the series-arm circuit and a ground potential GND. Each of the series-arm resonator units S1 to S5 and the parallel-arm resonator units P1 to P4 includes at least one acoustic wave resonator. In the example in FIG. 1, each of the series-arm resonator units S1 and S5 and the parallel-arm resonator units P1 to P4 includes one acoustic wave resonator, and each of the series-arm resonator units S2 to S4 includes two acoustic wave resonators. However, the number of acoustic wave resonators included in each resonator unit is not limited to this case, and may be appropriately selected in accordance with the characteristics of the filter. For example, a surface acoustic wave (SAW) resonator or a bulk acoustic wave (BAW) resonator may be used as each acoustic wave resonator.

(24) The parallel-arm resonator unit P1 includes a first end connected to a node between the series-arm resonator unit S1 and the series-arm resonator unit S2, and includes a second end connected to the ground potential GND via an inductor L1. The parallel-arm resonator unit P2 includes a first end connected to a node between the series-arm resonator unit S2 and the series-arm resonator unit S3, and includes a second end connected the ground potential GND via the inductor L1 as in the

parallel-arm resonator unit **P1**. The parallel-arm resonator unit **P3** includes a first end connected to a node between the series-arm resonator unit **S3** and the series-arm resonator unit **S4**, and includes a second end connected the ground potential GND via the inductor **L1** as in the cases of the parallel-arm resonator units **P1** and **P2**. The parallel-arm resonator unit **P4** includes a first end connected to a node between the series-arm resonator unit **S4** and the series-arm resonator unit **S5**, and includes a second end connected to the ground potential GND via an inductor **L2**.

(25) The reception filter **200** is a ladder filter connected between the antenna terminal **T1** and a reception terminal **T3**. The filter **200** filters a signal received at the antenna ANT, and outputs the signal from the reception terminal **T3**. The filter **200** allows signals in a pass band **BW2** (second pass band) to pass therethrough. The pass band **BW2** of the filter **200** is different from the pass band **BW1** of the filter **100**. The filter **200** is connected to the antenna terminal **T1** via an impedance matching inductor **L11**.

(26) The filter **200** includes a series-arm circuit including series-arm resonator units **S11** to **S14** serially connected between the inductor **L11** and the reception terminal **T3**, and a series-arm circuit including parallel-arm resonator units **P11** to **P14** connected between the series-arm circuit and the ground potential GND. Each of the series-arm resonator units **S11** to **S14** and the parallel-arm resonator units **P11** to **P14** includes at least one acoustic wave resonator. As in the case of the filter **100**, also in the filter **200**, the number of acoustic wave resonators included in each resonator unit is not limited to the case in FIG. 1, and may be appropriately selected in accordance with the characteristics of the filter. For example, a SAW resonator or a BAW resonator may be used as each acoustic wave resonator.

(27) The parallel-arm resonator unit **P11** includes a first end connected to a node between the inductor **L11** and the series-arm resonator unit **S11**, and includes a second end connected to the ground potential GND. The parallel-arm resonator unit **P12** includes a first end connected to a node between the series-arm resonator unit **S11** and the series-arm resonator unit **S12**, and includes a second end connected to the ground potential GND. The parallel-arm resonator unit **P13** includes a first end connected to a node between the series-arm resonator unit **S12** and the series-arm resonator unit **S13**, and includes a second end connected to the ground potential GND. The parallel-arm resonator unit **P14** includes a first end connected to a node between the series-arm resonator unit **S13** and the series-arm resonator unit **S14**, and includes a second end connected to the ground potential GND.

(28) The filter **100** corresponds to a “first filter”, and the filter **200** corresponds to a “second filter”.

(29) FIG. 2 is a cross-sectional view of the multiplexer **10** in FIG. 1. With reference to FIG. 2, the multiplexer **10** includes, in addition to the filters **100** and **200**, a mount board **50**, a resin layer **60**, and an electrically conductive layer **70**. An example where the filters **100** and **200** include SAW resonators used as acoustic resonators is described as an example, below.

(30) The filters **100** and **200** have a wafer level package (WLP) structure, and are mounted on the mount board **50** while being adjacent to each other. In the description below, a thickness direction of the mount board **50** is defined as a Z-axis, and, in an in-plane direction of the mount board **50**, a direction in which the filters **100** and **200** are adjacent to each other is defined as an X-axis direction. In the example in FIG. 2, the filter **100** is spaced apart from the filter **200** in a positive X-axis direction. A positive Z-axis direction may be referred to as a direction toward an upper surface, and a negative Z-axis direction may be referred to as a direction toward a lower surface.

(31) The mount board **50** is, for example, a monolayer or multilayer board made of resin such as epoxy or polyimide. Although not illustrated, the mount board **50** includes connection terminals provided on the upper surface and the lower surface, and a wiring pattern provided in the board and/or on the surface of the board.

(32) The filter **100** includes a piezoelectric substrate **110**, a support layer **120**, a cover layer **130**, a functional device **140**, a columnar electrode **150**, and a wiring pattern **160**.

(33) As illustrated in FIG. 3, the piezoelectric substrate **110** includes a base board **112**, a

piezoelectric layer **111** disposed on the lower surface of the base board **112**, and an intermediate layer **113** disposed between the piezoelectric layer **111** and the base board **112**.

(34) The piezoelectric layer **111** is made of, for example, a piezoelectric single-crystal material such as lithium tantalate (LiTaO.sub.3 or LT), lithium niobate (LiNbO.sub.3 or LN), alumina, silicon (Si), or sapphire, or a piezoelectric layered material including LiTaO.sub.3 or LiNbO.sub.3. The base board **112** is made of, for example, silicon (Si), lithium tantalate (LT), or lithium niobate (LN).

(35) The intermediate layer **113** includes a low-velocity layer **1131** and a high-velocity layer **1132**. The low-velocity layer **1131** and the high-velocity layer **1132** are arranged in order of the high-velocity layer **1132** and the low-velocity layer **1131** from adjacent to the base board **112** towards the piezoelectric layer **111**.

(36) The low-velocity layer **1131** is made of a material that allows a bulk wave to propagate therethrough at a lower velocity than a bulk wave that propagates through the piezoelectric layer **111**. The low-velocity layer **1131** is made of, for example, a dielectric such as a silicon dioxide (SiO.sub.2), glass, a silicon oxynitride, or a tantalum oxide, or a compound obtained by adding fluorine, carbon, or boron to a silicon dioxide.

(37) The high-velocity layer **1132** is made of a material that allows a bulk wave to propagate therethrough at a higher velocity than an acoustic wave that propagates through the piezoelectric layer **111**. The high-velocity layer **1132** is made of, for example, a material such as a silicon nitride (SiN), an aluminum nitride, an aluminum oxide (alumina), a silicon oxynitride, a silicon carbide, diamond-like carbon (DLC), or diamond.

(38) When the low-velocity layer **1131** and the high-velocity layer **1132** are disposed between the piezoelectric layer **111** and the base board **112**, the low-velocity layer **1131** and the high-velocity layer **1132** define and function as a reflecting layer (a mirror layer). Specifically, a surface acoustic wave that leaks from the piezoelectric layer **111** towards the base board **112** is reflected by the high-velocity layer **1132** due to the difference in propagation velocity, and confined in the low-velocity layer **1131**. Thus, the intermediate layer **113** reduces a loss of acoustic energy of the propagated surface acoustic wave, and thus can efficiently propagate the surface acoustic wave.

(39) Although FIG. 3 illustrates a case where the low-velocity layer **1131** and the high-velocity layer **1132** in the intermediate layer **113** are single layers, the intermediate layer **113** may include multiple low-velocity layers **1131** and multiple high-velocity layers **1132** disposed alternately with each other. The intermediate layer **113** may be omitted, and the piezoelectric substrate **110** may simply include the piezoelectric layer **111** and the base board **112**.

(40) The at least one functional device **140** is provided on the lower surface of the piezoelectric layer **111** of the piezoelectric substrate **110**. The functional device **140** includes a pair of interdigital transducer (IDT) electrodes made of an electrode material, for example, a single metal formed from at least one of aluminum, copper, silver, gold, titanium, tungsten, platinum, chromium, nickel, and molybdenum, or an alloy mainly including these metals. In the filter **100**, the piezoelectric substrate **110** and the IDT electrode form a surface acoustic wave (SAW) resonator.

(41) The support layer **120** is disposed on the lower surface of the piezoelectric substrate **110** and surrounds the periphery of the functional device **140**. The support layer **120** supports the cover layer **130** at a predetermined distance apart from the piezoelectric substrate **110** in the Z-axis direction. The support layer **120** and the cover layer **130** are made of an insulating resin mainly including, for example, epoxy, polyimide, acryl, or polyurethane. The piezoelectric substrate **110**, the support layer **120**, and the cover layer **130** define a hollow space **180**. The functional device **140** is disposed in the hollow space **180**.

(42) The wiring pattern **160** is formed on the lower surface of the piezoelectric substrate **110** to electrically connect the functional devices **140** to each other, and to electrically connect the functional devices **140** and the columnar electrode **150** (via) to each other.

(43) The columnar electrode **150** protrudes downward (negative Z-axis direction) from the lower

surface of the piezoelectric substrate **110**, and extends through the support layer **120** and the cover layer **130**. The columnar electrode **150** is connected to a connection terminal provided on the mount board **50** via an electrically conductive connector such as a solder bump **170**, for example. The columnar electrode **150** can electrically connect the functional device **140** and other electronic components on the mount board **50** to each other. The columnar electrode **150** and the wiring pattern **160** are made of, for example, an electrically conductive material such as aluminum, copper, silver, or gold.

(44) The filter **200** has the same or substantially the same structure as the filter **100**, and includes a piezoelectric substrate **210**, a support layer **220**, a cover layer **230**, a functional device **240**, a columnar electrode **250**, and a wiring pattern **260**. The functional device **240** is provided in a hollow space **280** defined by the piezoelectric substrate **210**, the support layer **220**, and the cover layer **230**. The filter **200** is connected to the mount board **50** via an electrically conductive connector such as a solder bump **270**, for example. The piezoelectric substrate **210** of the filter **200** includes a piezoelectric layer **211**, a base board **212**, and an intermediate layer **213**. Components in the filter **200** are the same or substantially the same as the corresponding components in the filter **100**, and thus are not repeatedly described in detail.

(45) The filters **100** and **200** mounted on the mount board **50** are sealed with the resin layer **60**. The resin layer **60** is made of, for example, a material obtained by mixing an inorganic filler such as metal into a material such as silicon compound, epoxy resin, silicone resin, fluorine resin, or acrylic resin. The electrically conductive layer **70** is disposed to cover the resin layer **60**.

(46) The electrically conductive layer **70** is made of, for example, a metal material having electric conductivity and high heat conductivity, such as aluminum, copper, silver, or gold. The electrically conductive layer **70** is connected to the ground potential at the mount board **50**. The electrically conductive layer **70** defines and functions as a shield to block leakage of electromagnetic noise generated in the multiplexer **10** to the outside of the device, and to block intrusion of electromagnetic noise into the multiplexer **10** from outside the device.

(47) As illustrated in FIG. **1**, in the multiplexer **10** according to the first preferred embodiment, the filter **100** is used as a transmission filter, and the filter **200** is used as a reception filter.

Transmission signals are generally transmitted using a power amplifier with power higher than the power used for reception signals to radiate radio waves as far as possible. Thus, the filter **100** consumes more power than the filter **200**, and generates more heat than the filter **200**.

(48) Therefore, in the multiplexer **10**, the upper surface of the piezoelectric substrate **110** in the filter **100** is exposed through the resin layer **60** to be in direct contact with the electrically conductive layer **70**. As described above, the electrically conductive layer **70** is made of a metal material with high heat conductivity. Thus, the electrically conductive layer **70** defines and functions as a heat sink of the filter **100**, and the electrically conductive layer **70** dissipates and radiates heat generated at the filter **100**.

(49) On the other hand, in the multiplexer **10** according to the first preferred embodiment, the piezoelectric substrate **210** of the filter **200** is thinner than the piezoelectric substrate **110** of the filter **100**, and not in contact with the electrically conductive layer **70**. As described above, the electrically conductive layer **70** defines and functions as a heat sink of the filter **100**, and has high heat conductivity. Thus, as in a multiplexer **10** # according to a comparative example in FIG. **4**, when a piezoelectric substrate **210** # of a filter **200** # is disposed to be in contact with the electrically conductive layer **70**, heat is transferred from the filter **100** to the filter **200** # via the electrically conductive layer **70**. Thus, heat from the filter **100** may raise the temperature of the filter **200** #, and may degrade filter characteristics of the filter **200** #.

(50) For example, depending on the material of the piezoelectric substrate, a temperature rise of the substrate may shift the pass band of the filter. When temperature characteristics of frequency (TCF) is positive (TCF>0), a temperature rise shifts the frequency band to higher frequencies. In contrast, when TCF is negative (TCF<0), a temperature rise shifts the frequency band to lower frequencies.

(51) In a structure such as in a multiplexer where filters having different pass bands are arranged adjacent to each other, and the two pass bands are approximate to each other, a shift of the pass band of one of the filters toward the pass band of the other filter may lower the impedance between the filters and degrade isolation. This may degrade the filter insertion loss. In the present disclosure, an increase of a such loss that occurs between filters adjacent to each other is referred to as a “parallel-arrangement loss”.

(52) In the multiplexer **10** according to the first preferred embodiment, the transmission filter **100** that generates a relatively large amount of heat includes the piezoelectric substrate **110** that is thicker (greater in the dimension in the Z-axis direction) and in direct contact with the electrically conductive layer **70**. Thus, the filter **100** improves the heat dissipation effect, and can reduce heat transferred via the resin layer **60**. In addition, the reception filter **200** that generates a relatively small amount of heat includes the piezoelectric substrate **210** that is thinner and not in contact with the electrically conductive layer **70**. Thus, the filter **200** reduces heat transferred via the electrically conductive layer **70**. This structure reduces heat of the filter **100** transferred to the filter **200**, and thus can reduce a parallel-arrangement loss resulting from the filter **100** and the filter **200** being arranged adjacent to each other.

(53) FIG. 5 is a diagram of an example of a temperature distribution map when the multiplexer **10** illustrated in FIG. 1 operates. In FIG. 5, the multiplexer **10** is disposed on a motherboard **300**, and a temperature distribution is indicated with contour lines. In FIG. 5, the temperature is highest at the filter **100**, and the temperature gradually lowers in a direction indicated with arrow **AR0**.

(54) As illustrated in FIG. 5, the temperature at the upper surface of the resin layer **60** facing the filter **200** is raised by the electrically conductive layer **70** further than the temperature around the filter **200**. FIG. 5 reveals that reduction of the thickness of the piezoelectric substrate **210** of the filter **200** reduces heat transferred to the piezoelectric substrate **210** via the electrically conductive layer **70**.

(55) FIG. 6 is a graph illustrating effects of the difference between the material of the piezoelectric substrates **110** and **210** and the difference of the contact state between the piezoelectric substrates **110** and **210** and the electrically conductive layer **70** on the temperature of the filter **200**. In FIG. 6, a solid line **LN1** indicates a case where the material of the piezoelectric substrate **110** is silicon (Si), the material of the piezoelectric substrate **210** is lithium niobate (LN), and the piezoelectric substrate **210** is not in contact with the electrically conductive layer **70**. A broken line **LN2** indicates a case where the material of both the piezoelectric substrates **110** and **210** is silicon, and the piezoelectric substrate **210** is not in contact with the electrically conductive layer **70**. A dot-and-dash line **LN3** indicates a case where the material of the two piezoelectric substrates is silicon, and the piezoelectric substrate **210** is in contact with the electrically conductive layer **70** (that is, a case of a comparative example in FIG. 4). A two-dot chain line **LN4** indicates a case where the material of both the piezoelectric substrates **110** and **210** is silicon, and the piezoelectric substrate **210** is not in contact with the electrically conductive layer **70**.

(56) In FIG. 6, an axis of abscissa indicates a distance in the X-axis direction from an outer end portion of the filter **200** (coordinate axis **a** in FIG. 2), and an axis of ordinate indicates a relative temperature difference with respect to the dot-and-dash line **LN3**.

(57) With reference to FIG. 6, in the cases of the first preferred embodiment where the piezoelectric substrate **210** is not in contact with the electrically conductive layer **70** (lines **LN1**, **LN2**, and **LN4**), the temperature of the piezoelectric substrate **210** is lower than that in the comparative example (line **LN3**). When lithium niobate is used as the material of the piezoelectric substrate **210**, the temperature gradient in the X-axis direction is greater than when silicon is used as the material of the piezoelectric substrate **210**. Thus, when a material having a great thermal resistance value such as, for example, lithium niobate is used as the material of the piezoelectric substrate **210**, and when the thickness of the piezoelectric substrate **210** is small, the characteristics of the piezoelectric substrate **210** may degrade also depending on the positions of the acoustic wave resonators in the

piezoelectric substrate **210**.

(58) Effects of Heat on Filter Characteristics

(59) With reference to FIG. 7, the reason why a loss caused in adjacent filters (parallel-arrangement loss) changes in response to a temperature rise is described. In FIG. 6, a case where the pass band BW1 of the transmission filter **100** (TX) is lower than the pass band BW2 of the reception filter **200** (RX) ($BW1 < BW2$) is described as an example.

(60) An upper section in FIG. 7 is a graph comparing insertion losses (broken lines LN11 and LN21) of the filters **100** and **200** defined by single bodies and insertion losses (solid lines LN10 and LN20) of the filters **100** and **200** built in the multiplexer **10**. As illustrated in the upper section in FIG. 7, when the filters **100** and **200** are connected to the common terminal as a multiplexer, insertion losses (parallel-arrangement losses) of both filters increase (arrow AR1). The parallel-arrangement losses are caused because the impedance of the adjacent filter cannot be completely infinite when viewed from the common terminal, and thus a subset of transmission signals leaks to the reception filter **200** or a subset of reception signals leaks to the transmission filter **100**.

(61) In this state, when the temperature of the filter **200** rises, the pass band BW2 of the filter **200** may change due to the temperature characteristics of the filter **200**. For example, when the pass band BW2 of the filter **200** lowers in response to a temperature rise ($TCF < 0$) (dot-and-dash line LN22), the pass band of the filter **200** becomes closer to the pass band BW1 of the filter **100**. Thus, isolation between the filter **100** and the filter **200** degrades, and the insertion loss of the filter **100** in the higher frequencies increases as indicated with an arrow AR2 (dot-and-dash line LN12).

Specifically, the temperature rise of the filter **200** increases the parallel-arrangement loss.

(62) On the other hand, when the pass band BW2 of the filter **200** rises in response to a temperature rise ($TCF > 0$) (dot-and-dash line LN23), the pass band BW2 of the filter **200** is spaced further apart from the pass band BW1 of the filter **100**. In this case, isolation between the filter **100** and the filter **200** is improved, and thus the insertion loss of the filter **100** in the higher frequencies decreases as indicated with an arrow AR3 (broken line LN13). Specifically, when $TCF > 0$, the parallel-arrangement loss decreases in response to a temperature rise.

(63) Conversely, when the pass band BW1 of the filter **100** is higher than the pass band BW2 of the filter **200** ($BW1 > BW2$), the parallel-arrangement loss decreases in response to a temperature rise when $TCF < 0$, and the parallel-arrangement loss increases in response to a temperature rise when $TCF > 0$.

(64) Hereafter, four cases that differ in the width relationship and TCF between the pass bands of the filters **100** and **200** are described in detail in relation to a change of a parallel-arrangement loss in response to a temperature rise and an arrangement of the corresponding acoustic wave resonators.

(65) Case 1

(66) FIG. 8 is a diagram describing a parallel-arrangement loss in case 1. Case 1 is a case where the pass band BW1 of the filter **100** is lower than the pass band BW2 of the filter **200** ($BW1 < BW2$), and the pass band BW2 of the filter **200** lowers together in response to a temperature rise ($TCF < 0$).

(67) In this case, as illustrated in the upper section in FIG. 8, the pass band BW2 of the filter **200** shifts to lower frequencies from a solid line LN40 to a broken line LN41 (arrow AR20). Thus, the insertion loss of the filter **100** in higher frequencies increases from a solid line LN30 to a broken line LN31 (arrow AR10).

(68) As illustrated in FIG. 1, the filter **200** is a ladder filter, for example. Thus, generally, attenuation poles in lower frequencies are provided by the parallel-arm resonator units P1 to P4. Thus, in case 1, to reduce an increase of the parallel-arrangement loss in response to a temperature rise, temperature rises of the parallel-arm resonator units are preferably reduced as much as possible, and attenuation poles of the pass band BW2 in lower frequencies are preferably shifted to higher frequencies. Thus, in case 1, as illustrated in FIG. 9, in the filter **200**, the parallel-arm resonator units P1 to P4 are spaced away from the filter **100** to reduce temperature rises of the

parallel-arm resonator units **P1** to **P4**.

(69) As illustrated in FIG. 9, in the filter **200**, with respect to a virtual line **CL1** that is perpendicular or substantially perpendicular to a direction in which the filters **100** and **200** are arranged, that is, perpendicular or substantially perpendicular to the X-axis direction (first direction) and that passes through the center of the piezoelectric substrate **210** of the filter **200**, a region closer to the filter **100** is defined as a region **RG1**, and a region farther from the filter **100** is defined as a region **RG2**. At this time, the parallel-arm resonator units **P1** to **P4** are arranged to allow a sum total **SM2** of the areas within the region **RG2** where the acoustic wave resonators included in the parallel-arm resonator units **P1** to **P4** are provided to be greater than a sum total **SM1** of the areas within the region **RG1** where the acoustic wave resonators included in the parallel-arm resonator units **P1** to **P4** are provided ($SM1 < SM2$).

(70) When the parallel-arm resonator units **P1** to **P4** are arranged in this manner, as illustrated in the lower section in FIG. 8, lower frequencies in the pass band **BW2** of the filter **200** shift to higher frequencies as indicated with a dot-and-dash line **LN42** (arrow **AR21**). Thus, isolation between the filter **100** and the filter **200** is improved, and an insertion loss of the pass band **BW1** of the filter **100** in higher frequencies can be reduced as indicated with a dot-and-dash line **LN32** (arrow **AR11**).

(71) In the above and the following description, as illustrated in FIG. 10A, “the area of a region where each acoustic wave resonator is provided” indicates the area of a cross region **SQ1** of electrode fingers in the IDT electrode **IDT** included in the acoustic wave resonator. Here, the cross region refers to an area where the multiple electrode fingers included in the IDT electrode overlap when the IDT electrode is viewed from a propagation direction of the acoustic wave. For example, in FIG. 10A, when the IDT electrode is viewed in the propagation direction of the acoustic wave, the area of the cross region is expressed by a product of **L1** and **L2** where **L1** denotes the length in which the electrode fingers extend, and **L2** denotes the distance between the outermost electrode fingers in the IDT electrode in the area where the multiple electrode fingers overlap.

(72) In the case of a longitudinally coupled resonator, the area of the cross region refers to the area of the cross region **SQ2** in FIG. 10B. The cross region in this case indicates an area where multiple electrode fingers included in the IDT electrodes **IDT1** to **IDT3** of the longitudinally coupled resonator overlap. For example, in FIG. 10B, when the IDT electrodes **IDT1** to **IDT3** are viewed in the propagation direction of the acoustic waves, the area of the cross region is expressed by a product of **L3** and **L4** where **L3** denotes the length in which the electrode fingers extend, and **L4** denotes the distance between the outermost electrode fingers in the IDT electrodes of the longitudinally coupled resonator in the area where the multiple electrode fingers overlap.

(73) Case 2

(74) FIG. 11 is a diagram describing a parallel-arrangement loss in case 2. Case 2 is a case where the pass band **BW1** of the filter **100** is lower than the pass band **BW2** of the filter **200** ($BW1 < BW2$) and the pass band **BW2** of the filter **200** shifts higher together in response to a temperature rise ($TCF > 0$). Specifically, in case 2, a temperature rise of the filter **200** expands a gap between the pass band of the filter **100** and the pass band of the filter **200**, and thus improves isolation.

(75) In this case, as illustrated in the upper section in FIG. 11, the pass band **BW2** of the filter **200** shifts to higher frequencies from a solid line **LN60** to a broken line **LN61** (arrow **AR40**). Thus, the insertion loss of the filter **100** in higher frequencies decreases from a solid line **LN50** to a broken line **LN51** (arrow **AR30**).

(76) As described also in case 1, in the ladder filter **200**, attenuation poles of the lower frequencies are provided by the parallel-arm resonator units **P1** to **P4**. In case 2, a temperature rise of the filter **200** reduces the parallel-arrangement loss. Thus, the temperatures of the parallel-arm resonator units are preferably raised as much as possible to shift the attenuation poles of the lower frequencies in the pass band **BW2** to further higher frequencies. Thus, in case 2, as illustrated in FIG. 12, the parallel-arm resonator units **P1** to **P4** in the filter **200** are arranged near the filter **100** to

promote temperature rises of the parallel-arm resonator units P1 to P4.

(77) Specifically, as in FIG. 12, in the filter 200, the parallel-arm resonator units P1 to P4 are arranged to allow a sum total SM1 of the areas within the region RG1 where the acoustic wave resonators included in the parallel-arm resonator units P1 to P4 are greater than a sum total SM2 of the areas within the region RG2 where the acoustic wave resonators included in the parallel-arm resonator units P1 to P4 are provided ($SM1 > SM2$).

(78) As illustrated in the lower section in FIG. 11, when the parallel-arm resonator units P1 to P4 are arranged in this manner, the lower frequencies in the pass band BW2 of the filter 200 shift to further higher frequencies as indicated with a dot-and-dash line LN62 (arrow AR41). Thus, isolation between the filter 100 and the filter 200 is further improved, and the insertion loss of the pass band BW1 of the filter 100 in higher frequencies can be further improved as indicated with a dot-and-dash line LN52 (arrow AR31).

(79) Case 3

(80) FIG. 13 is a diagram describing a parallel-arrangement loss in case 3. Case 3 is a case where the pass band BW1 of the filter 100 is higher than the pass band BW2 of the filter 200

($BW1 > BW2$), and the pass band BW2 of the filter 200 shifts higher together with a temperature rise ($TCF > 0$). Specifically, in case 3, a temperature rise of the filter 200 reduces a gap between the pass band of the filter 100 and the pass band of the filter 200, and thus degrades isolation.

(81) In this case, as illustrated in the upper section in FIG. 13, the pass band BW2 of the filter 200 shifts to higher frequencies from a solid line LN80 to a broken line LN81 (arrow AR60). Thus, an insertion loss of the filter 100 in lower frequencies increases from a solid line LN70 to a broken line LN71 (arrow AR50).

(82) In the ladder filter 200, attenuation poles of the higher frequencies are generally provided by the series-arm resonator units S1 to S5. In case 3, a temperature rise of the filter 200 increases the parallel-arrangement loss. Thus, temperature rises of the series-arm resonator units are preferably reduced as much as possible, and the attenuation poles of the higher frequencies in the pass band BW2 are preferably shifted to lower frequencies. Thus, in case 3, as illustrated in FIG. 14, the series-arm resonator units S1 to S5 in the filter 200 are spaced away from the filter 100 to reduce temperature rises of the series-arm resonator units S1 to S5.

(83) Specifically, as illustrated in FIG. 14, in the filter 200, the series-arm resonator units S1 to S5 are arranged to allow a sum total SM4 of the areas within the region RG2 where the acoustic wave resonators included in the series-arm resonator units S1 to S5 are greater than a sum total SM3 of the areas within the region RG1 where the acoustic wave resonators included in the series-arm resonator units S1 to S5 are provided ($SM3 < SM4$).

(84) When the series-arm resonator units S1 to S5 are arranged in this manner, as illustrated in the lower section in FIG. 13, higher frequencies in the pass band BW2 of the filter 200 shift to lower frequencies as indicated with a dot-and-dash line LN82 (arrow AR61). Thus, isolation between the filter 100 and the filter 200 is improved, and the insertion loss of the pass band BW1 of the filter 100 in lower frequencies can be improved as indicated with a dot-and-dash line LN72 (arrow AR51).

(85) Case 4

(86) FIG. 15 is a diagram describing a parallel-arrangement loss in case 4. Case 4 is a case where the pass band BW1 of the filter 100 is higher than the pass band BW2 of the filter 200

($BW1 > BW2$) and the pass band BW2 of the filter 200 lowers together with a temperature rise ($TCF < 0$). Specifically, in case 4, a temperature rise of the filter 200 expands a gap between the pass band of the filter 100 and the pass band of the filter 200, and thus improves isolation.

(87) In this case, as illustrated in the upper section in FIG. 15, the pass band BW2 of the filter 200 shifts to lower frequencies from a solid line LN100 to a broken line LN101 (arrow AR80). Thus, the insertion loss of the filter 100 in lower frequencies decreases from a solid line LN90 to a broken line LN91 (arrow AR70).

(88) As described also in case 3, in the ladder filter **200**, attenuation poles of the higher frequencies are provided by the series-arm resonator units **S1** to **S5**. In case 4, a temperature rise of the filter **200** reduces the parallel-arrangement loss. Thus, the temperatures of the series-arm resonator units are preferably raised as much as possible to shift the attenuation poles of the higher frequencies in the pass band **BW2** to further lower frequencies. Thus, in case 4, as illustrated in FIG. **16**, the series-arm resonator units **S1** to **S5** in the filter **200** are arranged near the filter **100** to promote temperature rises of the series-arm resonator units **S1** to **S5**.

(89) Specifically, as illustrated in FIG. **16**, in the filter **200**, the series-arm resonator units **S1** to **S5** are arranged to allow a sum total **SM3** of the areas within the region **RG1** where the acoustic wave resonators included in the series-arm resonator units **S1** to **S5** are greater than a sum total **SM4** of the areas within the region **RG2** where the acoustic wave resonators included in the series-arm resonator units **S1** to **S5** are provided ($SM3 > SM4$).

(90) As illustrated in the lower section in FIG. **15**, when the series-arm resonator units **S1** to **S5** are arranged in this manner, the higher frequencies in the pass band **BW2** of the filter **200** shift to further lower frequencies as indicated with a dot-and-dash line **LN102** (arrow **AR81**). Thus, isolation between the filter **100** and the filter **200** is further improved, and the insertion loss of the pass band **BW1** of the filter **100** in higher frequencies can be further improved as indicated with a dot-and-dash line **LN92** (arrow **AR71**).

(91) As described above, in the multiplexer according to the first preferred embodiment, one (first filter) of the two filters disposed adjacent to each other with a relatively small thermal resistance value is brought into contact with the electrically conductive layer, and the other one (second filter) of the two filters disposed adjacent to each other with a relatively great thermal resistance value is not in contact with the electrically conductive layer to reduce heat transfer between the adjacent filters. In addition, in the second filter, based on the temperature characteristics of frequency (TCF), the acoustic wave resonators are arranged to space the attenuation poles in the second filter located approximate to the pass band of the first filter apart from the pass band of the first filter. Thus, the parallel-arrangement loss caused in the first filter can be reduced by using the temperature gradient provided in the piezoelectric substrate by the thermal resistance. Thus, the degradation of the characteristics of the multiplexer (acoustic wave device) due to heat can be reduced.

(92) The first preferred embodiment has described a case where the transmission filter that generates a relatively large amount of heat is brought into contact with the electrically conductive layer. However, when the thermal resistance value of the transmission filter is greater than the thermal resistance value of the reception filter, the reception filter may be brought into contact with the electrically conductive layer, and the transmission filter may be not in contact with the electrically conductive layer.

(93) The piezoelectric substrates in the two filters may be made of different materials or the same material (for example, silicon). In that case, the thermal resistance value is adjusted by varying the thicknesses of the piezoelectric substrates.

(94) Instead of a combination of the transmission filter and the reception filter, both of the two filters may be transmission filters or reception filters as long as they have different pass bands. A combination of a transmission filter and a reception filter can be used as filters for a multiplexer, such as a pass band filter for transmitting signals between, for example, about 1710 MHz to about 1785 MHz and another pass band filter for receiving signals between, for example, about 1805 MHz to about 1880 MHz or a pass band filter for transmitting signals between, for example, about 2500 MHz to about 2570 MHz and another pass band filter for receiving signals between, for example, about 2620 MHz to about 2690 MHz. Two filters including transmission filters can be used as filters for a multiplexer, such as a pass band filter for transmitting signals between, for example, 1710 MHz to about 1785 MHz and another pass band filter for transmitting signals between, for example, about 1850 MHz to about 1915 MHz.

Second Preferred Embodiment

(95) In the first preferred embodiment, the two filters included in the multiplexer are both ladder acoustic wave filters. A second preferred embodiment of the present invention describes a case where a reception filter is a longitudinally coupled acoustic wave filter including longitudinally coupled resonators.

(96) FIG. 17 is a circuit configuration of a multiplexer 10A as an example of an acoustic wave device according to a second preferred embodiment of the present invention. With reference to FIG. 17, the multiplexer 10A includes an antenna terminal T1, and a transmission filter 100A and a reception filter 200A electrically connected to an antenna ANT at the antenna terminal T1.

(97) The transmission filter 100A is a ladder acoustic wave filter including a series-arm circuit including series-arm resonator units S21 to S25 serially connected between the antenna terminal T1 and the transmission terminal T2, and a parallel-arm circuit including parallel-arm resonator units P21 to P24 connected between the series-arm circuit and the ground potential GND. Each of the series-arm resonator units S21 to S25 and the parallel-arm resonator units P21 to P24 includes at least one acoustic wave resonator. In the example in FIG. 17, each of the series-arm resonator units S21 and S25 and the parallel-arm resonator units P21 to P24 includes one acoustic wave resonator, and each of the series-arm resonator units S22 to S24 includes two acoustic wave resonators. The number of acoustic wave resonators included in each resonator unit is not limited to this case, and may be appropriately selected in accordance with the characteristics of the filter. For example, a SAW resonator or a BAW resonator may be used as the acoustic wave resonator.

(98) The parallel-arm resonator unit P21 includes a first end connected to a node between the series-arm resonator unit S21 and the series-arm resonator unit S22, and includes a second end connected to the ground potential GND. The parallel-arm resonator unit P22 includes a first end connected to a node between the series-arm resonator unit S22 and the series-arm resonator unit S23, and includes a second end connected to the ground potential GND. The parallel-arm resonator unit P23 includes a first end connected to a node between the series-arm resonator unit S23 and the series-arm resonator unit S24, and includes a second end connected to the ground potential GND. The parallel-arm resonator unit P24 includes a first end connected to a node between the series-arm resonator unit S24 and the series-arm resonator unit S25, and includes a second end connected to the ground potential GND.

(99) The reception filter 200A includes series-arm resonator units S30 and S31 and an inductor L31. The series-arm resonator unit S30 and the series-arm resonator unit S31 are serially connected between the antenna terminal T1 and the reception terminal T3.

(100) The series-arm resonator unit S30 includes, for example, one acoustic wave resonator. The series-arm resonator unit S31 is a longitudinally coupled circuit including at least one longitudinally coupled resonator acoustic wave resonator. The series-arm resonator unit S31 includes three IDT electrodes IDT1 to IDT3 and two reflectors REF.

(101) The series-arm resonator unit S30 includes a first end connected to an antenna terminal T1 defining and functioning as a common terminal used together with the filter 100A. The IDT electrode IDT2 of the series-arm resonator unit S31 is connected between the second end of the series-arm resonator unit S30 and the ground potential GND.

(102) The IDT electrode IDT1 is disposed to face the first end of the IDT electrode IDT2 in the excitation direction, and the IDT electrode IDT3 is disposed to face the second end of the IDT electrode IDT2 in the excitation direction. The IDT electrode IDT1 and the IDT electrode IDT3 are connected in parallel between the reception terminal T3 and the ground potential GND. Reflectors REF are disposed to face the IDT electrodes IDT1 and IDT3 in the excitation direction opposite to the direction toward the IDT electrode IDT2.

(103) The inductor L1 is connected between the antenna terminal T1 and the ground potential GND. The inductor L1 defines and functions as an impedance matching inductor. The inductance of the inductor L1 is adjusted so that the impedance becomes open for high-frequency signals in the

pass band of the filter **100A** when the filter **200A** is viewed from the antenna terminal **T1**. This structure can reduce high-frequency signals in the pass band in the filter **100A** that pass toward the reception terminal **T3**.

(104) The cross-sectional view of the multiplexer **10A** according to the second preferred embodiment is the same or substantially the same as FIG. 2. The functional device **240** in FIG. 2 defines and functions as a longitudinally coupled resonator.

(105) Also in the multiplexer **10A**, the transmission filter **100A** that generates a relatively large amount of heat includes a piezoelectric substrate that is thicker and in direct contact with the electrically conductive layer. Thus, the filter **100A** improves the heat dissipation effect, and can reduce heat transferred via the resin layer. In addition, the reception filter **200A** that generates a relatively small amount of heat includes a piezoelectric substrate that is thinner and not in contact with the electrically conductive layer. Thus, the filter **200A** reduces heat transferred via the electrically conductive layer. This structure reduces heat of the filter **100A** transferred to the filter **200A**, and thus can reduce a parallel-arrangement loss resulting from the filter **100A** and the filter **200A** being arranged adjacent to each other.

(106) In the filter **200A**, the attenuation poles of the higher frequencies in the pass band and the attenuation poles of the lower frequencies in the pass band are both determined by the longitudinally coupled resonators. Thus, the parallel-arrangement loss in cases 1 and 3 in the first preferred embodiment can be reduced and the insertion loss in cases 2 and 4 in the first preferred embodiment can be further improved by arranging the acoustic wave resonators included in the longitudinally coupled circuit in the manner as specified in each case.

(107) More specifically, when the sum total of the areas within the region **RG1** in FIG. 9 where the acoustic wave resonators included in the longitudinally coupled circuit in the filter **200A** are provided is defined as **SM5** and the sum total of the areas within the region **RG2** in FIG. 9 where the acoustic wave resonators included in the longitudinally coupled circuit in the filter **200A** are provided is defined as **SM6**, the longitudinally coupled resonators are arranged to allow the sum total **SM6** of the areas within the region **RG2** to be greater than the sum total **SM5** of the areas within the region **RG1** in cases 1 and 3 ($SM5 < SM6$). In cases 2 and 4, the longitudinally coupled resonators are arranged to allow the sum total **SM5** of the areas within the region **RG1** to be greater than the sum total **SM6** of the areas within the region **RG2** ($SM5 > SM6$).

(108) While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. An acoustic wave device, comprising: a mount board; a first filter on the mount board and having a first pass band; a second filter on the mount board adjacent to the first filter in a first direction, and having a second pass band higher than the first pass band; a resin layer on the mount board and sealing the first filter and the second filter; and an electrically conductive layer covering the resin layer; wherein each of the first filter and the second filter includes: a piezoelectric substrate; and an acoustic wave resonator on the piezoelectric substrate; the first filter is in contact with the electrically conductive layer; the second filter is not in contact with the electrically conductive layer; the second filter is a ladder acoustic wave filter including at least one parallel-arm resonator and at least one series-arm resonator, or a longitudinally coupled acoustic wave filter including at least one longitudinally coupled resonator; a thermal resistance value of the piezoelectric substrate in the second filter in the first direction is greater than a thermal resistance value of the piezoelectric substrate in the first filter in the first direction; the second filter has characteristics to lower the second pass band in response to a temperature rise of the piezoelectric

substrate of the second filter; with respect to a virtual line perpendicular or substantially perpendicular to the first direction and passing through a center of the piezoelectric substrate of the second filter, a region of the piezoelectric substrate closer to the first filter is defined as a first region, and a region of the piezoelectric substrate farther from the first filter is defined as a second region; all of the at least one parallel-arm resonator included in the ladder acoustic wave filter is defined as a parallel-arm circuit, all of the at least one series-arm resonator is defined as a series-arm circuit, and all of the at least one longitudinally coupled resonator included in the longitudinally coupled acoustic wave filter is defined as a longitudinally coupled circuit; areas in the first region and the second region where the acoustic wave resonator included in the parallel-arm circuit or the longitudinally coupled circuit is provided are defined as a first area and a second area; and the second area is greater than the first area.

2. The acoustic wave device according to claim 1, wherein the piezoelectric substrate of the first filter includes silicon.

3. The acoustic wave device according to claim 1, wherein the acoustic wave resonator included in the second filter includes an interdigital transducer electrode.

4. The acoustic wave device according to claim 1, wherein the acoustic wave device is a multiplexer including the first filter and the second filter connected to a common terminal; the first filter defines and functions as a transmission filter to transmit a signal through the common terminal; and the second filter defines and functions as a reception filter to receive a signal received at the common terminal.

5. The acoustic wave device according to claim 1, wherein the electrically conductive layer includes a metal material.

6. The acoustic wave device according to claim 5, wherein the electrically conductive layer is at a ground potential.

7. The acoustic wave device according to claim 1, wherein the second filter includes a cover layer between the piezoelectric substrate of the second filter and the mount board, and the piezoelectric substrate and the cover layer define a hollow space; and the acoustic wave resonator included in the second filter is on the piezoelectric substrate of the second filter in the hollow space.

8. An acoustic wave device, comprising: a mount board; a first filter on the mount board and having a first pass band; a second filter on the mount board adjacent to the first filter in a first direction, and having a second pass band higher than the first pass band; a resin layer on the mount board and sealing the first filter and the second filter; and an electrically conductive layer covering the resin layer; wherein each of the first filter and the second filter includes: a piezoelectric substrate; and an acoustic wave resonator on the piezoelectric substrate; the first filter is in contact with the electrically conductive layer; the second filter is not in contact with the electrically conductive layer; the second filter is a ladder acoustic wave filter including at least one parallel-arm resonator and at least one series-arm resonator, or a longitudinally coupled acoustic wave filter including at least one longitudinally coupled resonator; a thermal resistance value of the piezoelectric substrate in the second filter in the first direction is greater than a thermal resistance value of the piezoelectric substrate in the first filter in the first direction; the second filter has characteristics to raise the second pass band in response to a temperature rise of the piezoelectric substrate of the second filter; with respect to a virtual line perpendicular or substantially perpendicular to the first direction and passing through a center of the piezoelectric substrate of the second filter, a region of the piezoelectric substrate closer to the first filter is defined as a first region, and a region of the piezoelectric substrate opposite to the region closer to the first filter is defined as a second region; all of the at least one parallel-arm resonator included in the ladder acoustic wave filter is defined as a parallel-arm circuit, all of the at least one series-arm resonator is defined as a series-arm circuit, and all of the at least one longitudinally coupled resonator included in the longitudinally coupled acoustic wave filter is defined as a longitudinally coupled circuit; areas in the first region and the second region where the acoustic wave resonator included in the

parallel-arm circuit or the longitudinally coupled circuit is provided are defined as a first area and a second area; and the first area is greater than the second area.

9. The acoustic wave device according to claim 8, wherein the piezoelectric substrate of the first filter includes silicon.

10. The acoustic wave device according to claim 8, wherein the acoustic wave resonator included in the second filter includes an interdigital transducer electrode.

11. The acoustic wave device according to claim 8, wherein the acoustic wave device is a multiplexer including the first filter and the second filter connected to a common terminal; the first filter defines and functions as a transmission filter to transmit a signal through the common terminal; and the second filter defines and functions as a reception filter to receive a signal received at the common terminal.

12. The acoustic wave device according to claim 8, wherein the electrically conductive layer includes a metal material.

13. The acoustic wave device according to claim 12, wherein the electrically conductive layer is at a ground potential.

14. The acoustic wave device according to claim 8, wherein the second filter includes a cover layer between the piezoelectric substrate of the second filter and the mount board, and the piezoelectric substrate and the cover layer define a hollow space; and the acoustic wave resonator included in the second filter is on the piezoelectric substrate of the second filter in the hollow space.

15. An acoustic wave device, comprising: a mount board; a first filter on the mount board and having a first pass band; a second filter on the mount board adjacent to the first filter in a first direction, and having a second pass band lower than the first pass band; a resin layer on the mount board and sealing the first filter and the second filter; and an electrically conductive layer covering the resin layer; wherein each of the first filter and the second filter includes: a piezoelectric substrate; and an acoustic wave resonator on the piezoelectric substrate; the first filter is in contact with the electrically conductive layer; the second filter is not in contact with the electrically conductive layer; the second filter is a ladder acoustic wave filter including at least one parallel-arm resonator and at least one series-arm resonator, or a longitudinally coupled acoustic wave filter including at least one longitudinally coupled resonator; a thermal resistance value of the piezoelectric substrate in the second filter in the first direction is greater than a thermal resistance value of the piezoelectric substrate in the first filter in the first direction; the second filter has characteristics to raise the second pass band in response to a temperature rise of the piezoelectric substrate of the second filter; with respect to a virtual line perpendicular or substantially perpendicular to the first direction and passing through a center of the piezoelectric substrate of the second filter, a region of the piezoelectric substrate closer to the first filter is defined as a first region, and a region of the piezoelectric substrate opposite to the region closer to the first filter is defined as a second region; all of the at least one parallel-arm resonator included in the ladder acoustic wave filter is defined as a parallel-arm circuit, all of the at least one series-arm resonator is defined as a series-arm circuit, and all of the at least one longitudinally coupled resonator included in the longitudinally coupled acoustic wave filter is defined as a longitudinally coupled circuit; areas in the first region and the second region where the acoustic wave resonator included in the series-arm circuit or the longitudinally coupled circuit is provided are defined as a third area and a fourth area; and the fourth area is greater than the third area.

16. The acoustic wave device according to claim 15, wherein the piezoelectric substrate of the first filter includes silicon.

17. The acoustic wave device according to claim 15, wherein the acoustic wave resonator included in the second filter includes an interdigital transducer electrode.

18. The acoustic wave device according to claim 15, wherein the acoustic wave device is a multiplexer including the first filter and the second filter connected to a common terminal; the first filter defines and functions as a transmission filter that transmits a signal through the common

terminal; and the second filter defines and functions as a reception filter that receives a signal received at the common terminal.

19. The acoustic wave device according to claim 15, wherein the electrically conductive layer includes a metal material.

20. The acoustic wave device according to claim 19, wherein the electrically conductive layer is at a ground potential.

21. The acoustic wave device according to claim 15, wherein the second filter includes a cover layer between the piezoelectric substrate of the second filter and the mount board, and the piezoelectric substrate and the cover layer define a hollow space; and the acoustic wave resonator included in the second filter is on the piezoelectric substrate of the second filter in the hollow space.

22. An acoustic wave device, comprising: a mount board; a first filter on the mount board and having a first pass band; a second filter on the mount board adjacent to the first filter in a first direction, and having a second pass band lower than the first pass band; a resin layer disposed on the mount board to seal the first filter and the second filter; and an electrically conductive layer covering the resin layer; wherein each of the first filter and the second filter includes: a piezoelectric substrate; and an acoustic wave resonator on the piezoelectric substrate; the first filter is in contact with the electrically conductive layer; the second filter is not in contact with the electrically conductive layer; the second filter is a ladder acoustic wave filter including at least one parallel-arm resonator and at least one series-arm resonator, or a longitudinally coupled acoustic wave filter including at least one longitudinally coupled resonator; a thermal resistance value of the piezoelectric substrate in the second filter in the first direction is greater than a thermal resistance value of the piezoelectric substrate in the first filter in the first direction; the second filter has characteristics to lower the second pass band in response to a temperature rise of the piezoelectric substrate of the second filter; with respect to a virtual line perpendicular or substantially perpendicular to the first direction and passing through a center of the piezoelectric substrate of the second filter, a region of the piezoelectric substrate closer to the first filter is defined as a first region, and a region of the piezoelectric substrate opposite to the region closer to the first filter is defined as a second region; all of the at least one parallel-arm resonator included in the ladder acoustic wave filter is defined as a parallel-arm circuit, all of the at least one series-arm resonator is defined as a series-arm circuit, and all of the at least one longitudinally coupled resonator included in the longitudinally coupled acoustic wave filter is defined as a longitudinally coupled circuit; areas in the first region and the second region where the acoustic wave resonator included in the series-arm circuit or the longitudinally coupled circuit is provided are defined as a third area and a fourth area; and the third area is greater than the fourth area.

23. The acoustic wave device according to claim 22, wherein the piezoelectric substrate of the first filter includes silicon.

24. The acoustic wave device according to claim 22, wherein the acoustic wave resonator included in the second filter includes an interdigital transducer electrode.

25. The acoustic wave device according to claim 22, wherein the acoustic wave device is a multiplexer including the first filter and the second filter connected to a common terminal; the first filter defines and functions as a transmission filter to transmit a signal through the common terminal; and the second filter defines and functions as a reception filter to receive a signal received at the common terminal.

26. The acoustic wave device according to claim 22, wherein the electrically conductive layer includes a metal material.

27. The acoustic wave device according to claim 26, wherein the electrically conductive layer is at a ground potential.

28. The acoustic wave device according to claim 22, wherein the second filter includes a cover layer between the piezoelectric substrate of the second filter and the mount board, and the

piezoelectric substrate and the cover layer define a hollow space; and the acoustic wave resonator included in the second filter is on the piezoelectric substrate of the second filter in the hollow space.
